

MICHAEL HAYES

VMware Cloud Foundation Architect

St. Cloud, FL | 401-854-6341 | mhayes@virtualcontrolllc.com | linkedin.com/in/mrmichael-hayes

SUMMARY

VMware Cloud Foundation architect with 25 years of hands-on enterprise virtualization experience, spanning design, deployment, validation and root-cause troubleshooting across the full VCF 9 stack: vSphere, vSAN (ESA and OSA), NSX, SDDC Manager, VCF Operations, VCF Operations for Logs and VCF Automation. Most recent engagement delivered a VCF Operations 9 monitoring design for a Fortune 100 energy utility through a Broadcom Premier Partner, issued as a client-facing High-Level Design. Previously sole architect and sole engineer for a complete VMware estate across a nine-year engagement, including full ownership of storage.

Documents design decisions as decision, justification and implication, traced to the requirement they serve and cited to Broadcom source. Independent practice work covers brownfield VCF recovery and inventory reconciliation, vpxd and cluster-store root-cause analysis, 9.0-to-9.1 upgrade planning, and a 100+ check automated VCF assessment platform. Author of a 60-page VCF 9 architecture design playbook in which every claim carries a Broadcom source and a verification date.

CERTIFICATIONS

- **VMware Cloud Foundation 9 Administrator**, current
- **VMware Certified Professional, VCF Architect (2V0-13.25)**, in progress.
- VMware Certified Professional, Data Center Virtualization (VCP-DCV), versions 3 through 6.5
- AWS Cloud Practitioner · AI Foundation Certificate and AI Mainframe Modernization (Kyndryl)

DESIGN CAPABILITY

Design method. Requirement traceability from business objective through technical requirement. Conceptual, logical and physical design layers. Separate registers for requirements, constraints, assumptions and risks. Design decisions recorded with justification and implication. Design validation criteria defined alongside the requirements.

VCF 9 architecture. Fleet, instance and domain taxonomy. Design blueprints and the underlying deployment models: cluster models (single-rack, Layer 2 and Layer 3 multi-rack, stretched), vSAN storage models, network fabric models, NSX management and Edge cluster models, Single Sign-On models, VCF Operations and VCF Automation deployment models, tenancy and consumption models.

AMPRS design. Availability (stretched clusters, fault domains, admission control sizing), manageability (vLCM images, VCF Operations fleet lifecycle), performance (vSAN network sizing, Network I/O Control shares), recoverability (RPO and RTO driven), security (certificate, identity and access design).

Engineering depth. PowerCLI, PowerShell, Python, Bash, SQL. REST API development (FastAPI / OpenAPI Swagger, JWT auth). Infrastructure as Code with VMware Aria Automation 8.18. NetApp ONTAP REST integration. Kubernetes via VKS / TKG, CNS and first class disks. Photon OS appliance deployment, systemd and cron service automation.

PROFESSIONAL EXPERIENCE

Entelligence (Broadcom Premier Partner) *contracted via Virtual Control LLC*
Senior VMware Cloud Foundation / VCF Operations Engineer · Apr 2026 to Jul 2026

Brought in to design the storage monitoring layer for a Fortune 100 energy utility's VCF 9 platform. Delivered a NetApp ONTAP to VCF Operations 9 integration, issued as a client-facing VMware by Broadcom High-Level Design with production build and configuration guides, per-team coordination hand-off packets, and a stakeholder Q&A brief.

- Designed the collection architecture: a custom management pack polling the NetApp ONTAP REST API for per-volume and per-aggregate capacity, performance (IOPS / latency / throughput) and storage-efficiency telemetry (49 metrics, 48 properties).
- Designed the analytics layer against stated operational requirements: single-pane dashboards, list views, schedulable reports, and super metrics for per-volume percent-free and efficiency ratio.
- Designed the alerting model: 6 alert types across 7 definitions covering latency, free space, IOPS anomaly, aggregate over-commitment, efficiency drop, and not-collecting states for volumes and aggregates.

Virtual Control LLC *founded 2016; the vehicle for the engagements above*
Principal Architect · 2016 to Present

Independent VMware practice. Runs a full nested VCF 9.0.1 environment (vSphere, vSAN, NSX, SDDC Manager, VCF Operations, Aria Automation) end-to-end for capability development, testing and client enablement.

Design artifacts

- **VCF 9 Architecture Design Playbook:** a 60-page design guide organised by decision dependency, covering the four decisions most expensive to reverse. Every claim carries a Broadcom source and a verification date.
- **VCF 9 Architect Engagement Handbook:** a 62-page engagement methodology covering assessment through education, high-level design and modernization, written so a delivery team can execute a design engagement consistently.

- **VCF HLD generator:** tooling that produces a first-draft High-Level Design from an engagement intake, validates the intake against published VCF limits for latency, MTU and host counts, and cites all 377 published VCF 9.0 design elements to source. Blocks output when the intake violates a published limit (27 automated tests).
- **VCF 9 design decision corpus:** 377 published requirements and recommendations indexed against a 6,346-page Broadcom documentation search platform.

Engineering and root-cause work

- Recovered a brownfield SDDC Manager host-inventory desync. Three ESXi hosts were left orphaned (status None, no cluster reference) after an out-of-band vCenter redeploy; root-caused the aborting domain guardrails, remediated a stray host VMkernel tag, and drove Sync Inventory to completion. Closed with no vendor support case.
- Root-caused a vCenter vpxd service crash to an ESX cluster-store (DKVS) reconfigure loop and half-joined replica (Broadcom KB 385346, resolved in vSphere 9.1); authored the RCA and stood up a scheduled read-only monitor that flags the escalation condition.
- Diagnosed a cluster-wide vSAN write-performance collapse: traced it to a stray vSAN-tagged VMkernel adapter injecting a dead unicast peer. Restored roughly 10x throughput.
- Built the VCF Health Check platform: 100+ automated checks across vCenter, NSX, SDDC Manager, vSAN, VCF Operations, Fleet / Lifecycle and Aria Automation, producing graded HTML / PDF / JSON reports with per-component scoring, heatmaps and historical trend analysis. Delivered as desktop GUI, headless CLI and REST API, and provisioned as a self-running Photon OS appliance via an Aria Automation cloud template.
- Authored verified design and operations references: VCF Single Sign-On integration with Active Directory including the connection-security decision and per-component role mapping, and a VCF 9.0-to-9.1 upgrade runbook covering prechecks, VCF Management Services deployment, License Server, and resource and HCL gating.

Thielsch Engineering (VMware Partner) *contracted via Virtual Control LLC; held concurrently with the staff roles below*
VMware Architect · Mar 2016 to Dec 2025

Sole architect and sole engineer for the entire VMware environment across a nine-year engagement, with end-to-end responsibility for the storage estate. Fully redesigned and supported the platform through every major release from 5.5 to 8.0 U3G, then into VCF 9.

- Deployed VCF 9 in production and lab: SDDC Manager, NSX, vSAN, ESXi 9.0, Fleet Management, VCF Operations and VCF Operations for Logs.
- Led an Azure migration and a vCenter consolidation improving efficiency by 30%; decommissioned EMC VNX arrays and migrated workloads to NetApp.
- Designed and supported VMware Site Recovery Manager for DR failover; implemented 10 Gb trunking reducing latency by 30%.

Kyndryl (Citizens Bank, True Value, HCSC, NYMTA, Toyota) *concurrent with the Thielsch engagement*
Performance & Capacity Engineer · Dec 2019 to Feb 2026

- Capacity and performance engineering across five enterprise accounts: predictive capacity models for resource planning and SLA compliance, tuning that cut system overhead 10% and MIPS consumption 15%, and PowerShell / ODBC automation for performance reporting.

Advantasure / BCBS *concurrent with the Thielsch engagement*
VMware & Cloud Senior Architect · 2016 to 2019

- VMware design, VM hardening and infrastructure modernization. vCenter consolidation reducing hardware footprint and support calls by 80%. Deployed vROps 6.6 / 7.0, improving virtual server performance 33%.

Johnson & Johnson (VMware Partner)
Senior Consultant / Technical Lead · 2015 to 2016

- Administered 8,000+ VMs across 8 global vCenters and 400+ ESXi hosts. Established performance baselines and optimization standards; mentored engineering teams on SDDC capability.

Citizens Bank
Senior Infrastructure Engineer, AVP · 2012 to 2015

- Co-founded and led the Virtualization Department. Tier 3 VMware SME for enterprise operations. Three Good Banking Awards and a Bank Credo Award

Earlier. Senior VMware and infrastructure roles at Hasbro, EMC, VMware, Genzyme and McKesson, 2000 to 2012.

PLATFORM SCOPE

Largest estate administered: 8,000+ VMs, 8 vCenters, 400+ ESXi hosts.

Longest single engagement: 9 years as sole architect and engineer, through 6 major vSphere releases.

Storage: vSAN ESA and OSA, NetApp ONTAP, Fibre Channel, NFS, VMFS, iSCSI, NVMe/TCP.

DR: Site Recovery Manager, Zerto, Veeam, EMC RecoverPoint.